

**A Report
on
ANALYSIS OF PLATE USING
TRIANGULAR PLANE STRESS ELEMENT WITH
DISTRIBUTED LOAD**



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CHAPTER-1

1. INTRODUCTION

1. Introduction:

The plane stress problem involves analyzing the deformation in x and y directions using triangular element where loading is distributed. This study employs the Finite Element Method(FEM) to solve the system of equations over a domain with specified boundary conditions. This study helps develop a deeper understanding of how thin plates behave when subjected distributed loads, such as in load-bearings, pressure vessel, and machine components.

CHAPTER-2

MATHEMATICAL MODEL

2.1 Governing Equation

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + f_x = 0$$

$$\frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + f_y = 0$$

2.2 Weak Formulation

The weak form of the governing equation is derived by multiplying the PDE with a weight function and integrating over the domain also we have considered no traction force.

Where δu and δv are the weight function.

$$\int_{\Omega} \delta u \left(\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + f_x \right) dx dy + \int_{\Omega} \delta v \left(\frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + f_y \right) dx dy = 0$$

Displacement field: $U = N * q_e$

Strain field: $\epsilon = B * q_e$

Stress field: $\sigma = C * B * q_e$

Integral Functional I :

$$I = - \int_{\Gamma} \delta u t_x ds - \int_{\Gamma} \delta v t_y ds + \int_{\Omega} \left((\sigma_{xx} \delta \epsilon_{xx} + \sigma_{yy} \delta \epsilon_{yy}) + \sigma_{xy} \delta v_{xy} \right) dx dy \\ - \int_{\Omega} (\delta u f_x + \delta v f_y) dx dy = 0$$

Elemental Equation :

$$\delta I = \int_{\Omega_e} \delta [de]^T [B]^T [D] [B] [de] dx dy - \int_{\Omega_e} \delta [de]^T [H]^T \begin{bmatrix} f_x \\ f_y \end{bmatrix} dx dy - \int_{\Gamma_e} \delta [de]^T [H]^T \begin{bmatrix} t_x \\ t_y \end{bmatrix} ds = 0$$

Stiffness Matrix :

$$[Ke] = \int_{\Omega} B^T C B = \pi r^2$$

CHAPTER-3

Results

3.1 Simulation Setup

- Plate dimensions: $L_x=5$ m, $L_y=1$ m
- Finite Element Mesh: 20 divisions along the length, 4 divisions along the height,
Total nodes: 105, Total triangular elements: 160
- Element Type: Linear 3-node triangular plane stress element
- Material Properties: Young's Modulus: $E=200$ GPa, Poisson Ratio: $\nu=0.30$,
Plane stress constitutive matrix computed accordingly.
- Loading Condition: A distributed horizontal load is applied along the right vertical edge:

$$P_{\text{total}} = 10^8 \text{ N}$$

Distributed uniformly from bottom to top:

$$q_0 = \frac{P_{\text{total}}}{L_y}$$

- Boundary Conditions:
Left edge: all $u_x=0$ (to prevent horizontal movement).
One corner node at (0,0): $u_y=0$ (to prevent rigid-body translation).
All other vertical DOFs are free.

1. Mesh Resolution Analysis

The analysis was performed using a mesh of 20 divisions along the length and 4 divisions along the height, resulting in total of 105 nodes and 160 triangular elements. This discretization provides a sufficiently fine resolution to capture the stress distribution produced by the applied distributed load. The mesh ensures stable numerical behavior and smooth stress contours across the plate.

2. Boundary Conditions Impact

The imposed boundary conditions-fixing all the horizontal displacements on the left edge and constraining the vertical displacement of a single corner node-ensure that the plate undergoes pure axial deformation without any rigid-body motion. These constraints create a stable reference frame, allowing the distributed load on the right edge to stretch the plate uniformly. As a result, the horizontal displacement increases smoothly from the fixed left edge toward the loaded boundary, while vertical displacements remain negligible, consistent with the expected behavior of a plate under axial loading.

3. Displacement Results

The computed nodal displacements show the classical axial deformation pattern horizontal distributed traction.

- Maximum horizontal displacement: $u_{x,max} = 2.5 \times 10^{-2}$
- Maximum vertical displacement: $u_{y,max} = 1.5 \times 10^{-3}$
- Deformed Shape Observation:
 - The right edge moves uniformly outward.
 - No bending or curvature appears in the deformation field, confirming pure axial loading.

4. Visualization Insights

The undeformed mesh plot confirms proper mesh generation and element connectivity. Deformed mesh shows smooth extension without artificial twisting or bending. Stress plots clearly reveal a directionally dominant axial stress.

5. Potential Engineering Applications

The study provides insights into the structural behavior of thin plate components subjected to distributed edge loads. The methodology can be extended to analyze and optimize a wide range of engineering structures such as machine plates, sheet-metal parts, automotive body sections, and aerospace skin components. The approach is also applicable in evaluating load-bearing elements in different areas where accurate prediction of stress and deformation in flat plate region is essential.

3.2 Plot of deformation of Plate

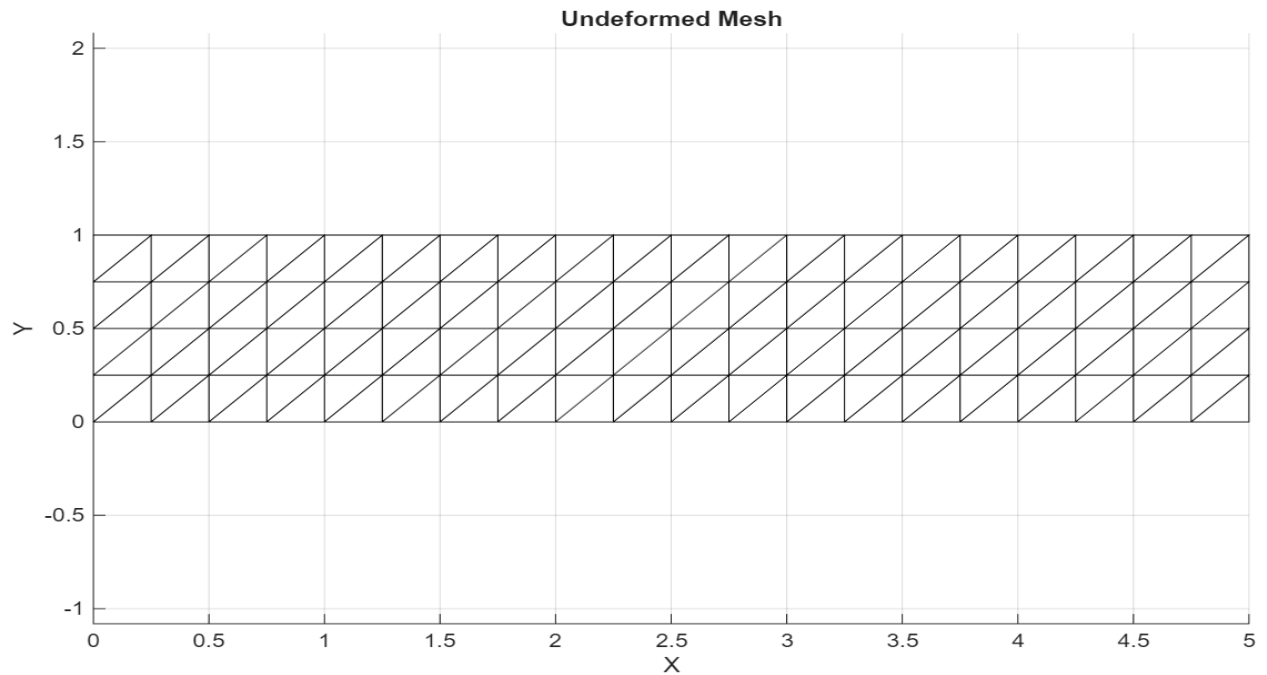


Fig. 1 Undeformed Mesh

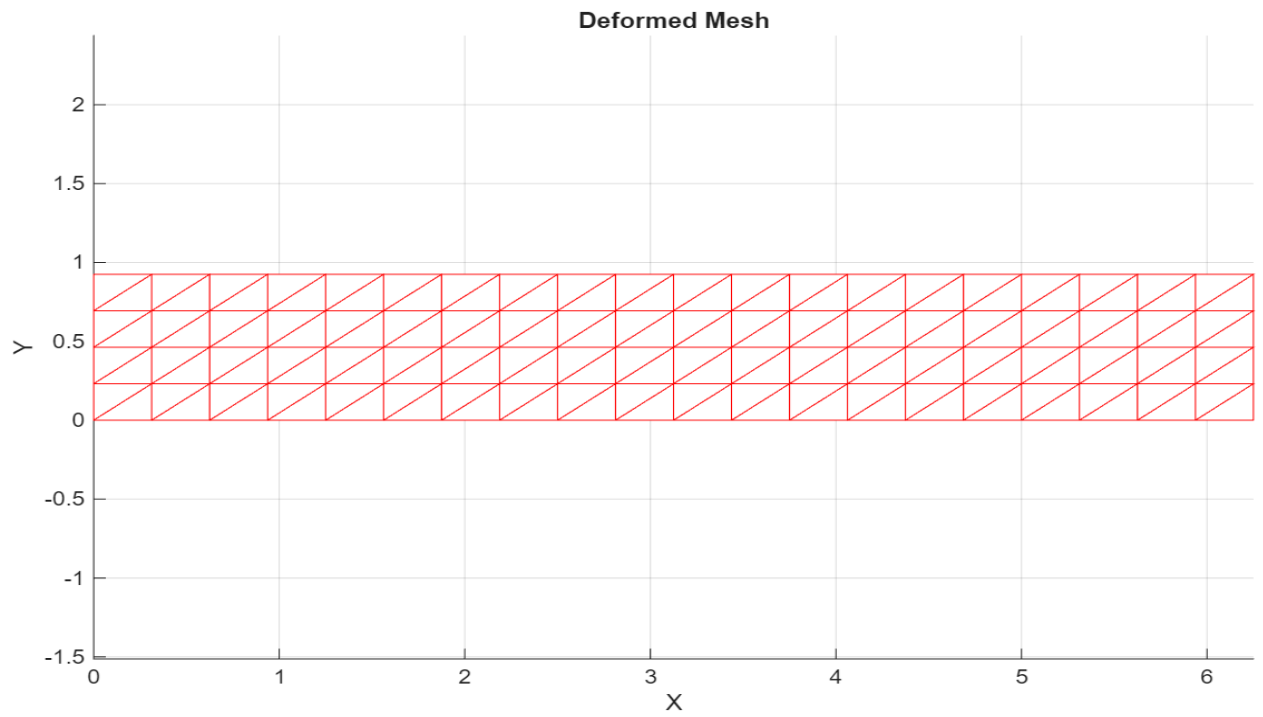


Fig. 2 Deformed Mesh

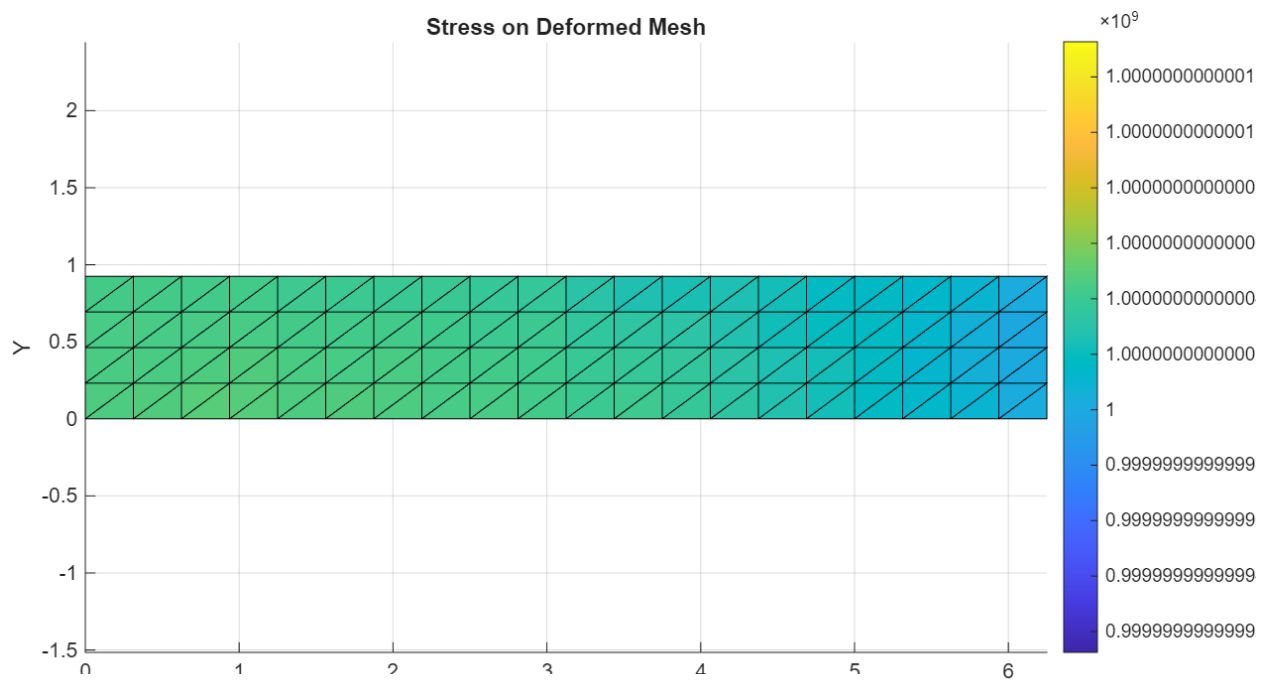


Fig. 3 Stress variation on Deformed Mesh

CHAPTER-4

4. CONCLUSION

Conclusion

This study demonstrates the application of the Finite Element Method for analyzing the structural response of a rectangular plate subjected to a distributed edge load using triangular plane stress elements. The analysis confirms that the imposed boundary conditions ensure pure axial deformation, while the distributed load produces a realistic stress distribution in accordance with plane stress theory. The study validates the robustness of FEM for handling structural problems involving distributed loads, offering useful insights for evaluating thin plate components in practical engineering applications.

CHAPTER-5

MATLAB CODE

```
% GEOMETRY
Lx = 5.0;      % length (m)
Ly = 1.0;      % height (m)

nx = 20;       % number of divisions in x
ny = 4;        % number of divisions in y

nn = (nx+1)*(ny+1); % number of nodes
nel = 2*nx*ny;     % 2 triangles per rectangle

% MATERIAL
E = 200e9;       % Young modulus (Pa)
nu = 0.30;       % Poisson ratio

C = E/(1-nu^2)*[ 1  nu      0;
                nu  1      0;
                0  0  (1-nu)/2 ]; % Plane stress constitutive matrix

% GENERATE NODE COORDINATES
node = zeros(nn,2);
count = 0;
for j = 0:ny
    for i = 0:nx
        count = count + 1;
        node(count,1) = Lx * i/nx;
        node(count,2) = Ly * j/ny;
    end
end

% GENERATE TRIANGULAR ELEMENT CONNECTIVITY
elem = zeros(nel,3);
e = 0;

for j = 1:ny
    for i = 1:nx
        n1 = (j-1)*(nx+1) + i;
        n2 = n1 + 1;
        n3 = n1 + (nx+1);
        n4 = n3 + 1;

        % Triangle 1 (n1, n2, n4)
        e = e + 1;
        elem(e,:) = [n1 n2 n4];

        % Triangle 2 (n1, n4, n3)
        e = e + 1;
        elem(e,:) = [n1 n4 n3];
    end
end
```

```

    end
end

% ASSEMBLY
K = zeros(2*nn);

for el = 1:nel
    nodes_el = elem(el,:);
    x1 = node(nodes_el(1),1);    y1 = node(nodes_el(1),2);
    x2 = node(nodes_el(2),1);    y2 = node(nodes_el(2),2);
    x3 = node(nodes_el(3),1);    y3 = node(nodes_el(3),2);

    % Area of triangular element
    A = 0.5 * det([1 x1 y1; 1 x2 y2; 1 x3 y3]);

    % Bi, Ci for shape functions
    b1 = y2 - y3;    c1 = x3 - x2;
    b2 = y3 - y1;    c2 = x1 - x3;
    b3 = y1 - y2;    c3 = x2 - x1;

    B = 1/(2*A)*[ b1    0    b2    0    b3    0;
                  0    c1    0    c2    0    c3;
                  c1    b1    c2    b2    c3    b3 ];

    ke = A * (B') * C * B;

    % Assembly into global K
    dofs = [2*nodes_el(1)-1 2*nodes_el(1) ...
            2*nodes_el(2)-1 2*nodes_el(2) ...
            2*nodes_el(3)-1 2*nodes_el(3)];

    K(dofs,dofs) = K(dofs,dofs) + ke;
end

% LOAD VECTOR: DISTRIBUTED LOAD ON RIGHT EDGE IN X
F = zeros(2*nn,1);

P_total = 10e8;          % total resultant in +x (N)
q0 = P_total / Ly;       % line load (N/m) along right edge

tol = 1e-12;

% nodes on right edge x = Lx
right_nodes = find(abs(node(:,1) - Lx) < tol);
[~, idx] = sort(node(right_nodes,2)); % bottom to top
right_nodes = right_nodes(idx);

for k = 1:length(right_nodes)-1
    n1 = right_nodes(k);
    n2 = right_nodes(k+1);

    y1 = node(n1,2);    y2 = node(n2,2);
    segLen = y2 - y1;

    Fx_seg = q0 * segLen; % total on this segment

```

```

F(2*n1 - 1) = F(2*n1 - 1) + Fx_seg/2;
F(2*n2 - 1) = F(2*n2 - 1) + Fx_seg/2;
end

% BOUNDARY CONDITIONS FOR PURE AXIAL LOADING
% All left-edge nodes: fix ux = 0, uy free
left_nodes = find(abs(node(:,1)) < tol);

bc_dofs = [];

% Fix ux on entire left edge
for k = 1:length(left_nodes)
    n = left_nodes(k);
    bc_dofs = [bc_dofs, 2*n - 1]; % ux
end

% Fix uy of ONE reference node to stop rigid-body motion in y
% choose bottom-left corner: x=0, y=0
corner_node = find(abs(node(:,1)) < tol & abs(node(:,2)) < tol);
corner_node = corner_node(1); % just in case multiple match
bc_dofs = [bc_dofs, 2*corner_node]; % uy at one node

bc_dofs = unique(bc_dofs);
all_dofs = 1:(2*nn);
active_dofs = setdiff(all_dofs, bc_dofs);

Kred = K(active_dofs, active_dofs);
Fred = F(active_dofs);

% SOLVE
U = zeros(2*nn,1);
U(active_dofs) = Kred \ Fred;

Ux = U(1:2:end);
Uy = U(2:2:end);

fprintf("Max horizontal displacement = %.6e m\n", max(abs(Ux)));
fprintf("Max vertical displacement = %.6e m\n", max(abs(Uy)));

% UNDEFORMED MESH
figure(1); hold on; axis equal; grid on;
title('Undeformed Mesh');
xlabel('X'); ylabel('Y');
for el = 1:nel
    nd = elem(el,:);
    fill(node(nd,1), node(nd,2), 'w', 'EdgeColor','k');
end

% DEFORMED MESH
scale = 50;
deformed = node + scale * [Ux Uy];

figure(2); hold on; axis equal; grid on;
title(['Deformed Mesh']);

```

```

xlabel('X'); ylabel('Y');
for el = 1:nel
    nd = elem(el,:);
    fill(deformed(nd,1), deformed(nd,2), 'w', 'EdgeColor','r');
end

% STRESS / STRAIN COMPUTATION
strain_elem = zeros(3, nel); % [exx; eyy; gxy]
stress_elem = zeros(3, nel); % [sxx; syy; txy]
sigma_vm = zeros(nel,1); % von Mises

for el = 1:nel
    nodes_el = elem(el,:);
    x1 = node(nodes_el(1),1); y1 = node(nodes_el(1),2);
    x2 = node(nodes_el(2),1); y2 = node(nodes_el(2),2);
    x3 = node(nodes_el(3),1); y3 = node(nodes_el(3),2);

    A = 0.5 * det([1 x1 y1; 1 x2 y2; 1 x3 y3]);

    b1 = y2 - y3; c1 = x3 - x2;
    b2 = y3 - y1; c2 = x1 - x3;
    b3 = y1 - y2; c3 = x2 - x1;

    B = 1/(2*A)*[ b1 0 b2 0 b3 0;
                  0 c1 0 c2 0 c3;
                  c1 b1 c2 b2 c3 b3 ];

    dofs = [2*nodes_el(1)-1 2*nodes_el(1) ...
            2*nodes_el(2)-1 2*nodes_el(2) ...
            2*nodes_el(3)-1 2*nodes_el(3)];
    Ue = U(dofs);

    eps_el = B * Ue;
    sig_el = C * eps_el;

    strain_elem(:,el) = eps_el;
    stress_elem(:,el) = sig_el;

    sxx = sig_el(1); syy = sig_el(2); txy = sig_el(3);
    sigma_vm(el) = sqrt(sxx^2 - sxx*syy + syy^2 + 3*txy^2);
end

% NODAL AVERAGING OF STRESSES
nodeStressSum = zeros(nn,3);
nodeStressCount = zeros(nn,1);

for el = 1:nel
    nodes_el = elem(el,:);
    sig_el = stress_elem(:,el)'; % [sxx syy txy]
    for a = 1:3
        n = nodes_el(a);
        nodeStressSum(n,:) = nodeStressSum(n,:) + sig_el;
        nodeStressCount(n,1) = nodeStressCount(n,1) + 1;
    end
end

```

```

nodeStress = nodeStressSum ./ max(nodeStressCount,1); % [sxx syy txy] at
nodes

% STRESS ON DEFORMED MESH
figure(3);
trisurf(elem, deformed(:,1), deformed(:,2), zeros(nn,1), nodeStress(:,1), ...
    'EdgeColor','k');
view(2); axis equal; grid on; colorbar;
title('Stress on Deformed Mesh');
xlabel('X'); ylabel('Y');

```

REFERENCES

- [1] J.N. Reddy, "An Introduction to the Finite Element Method", 2017 3rd Edition.
- [2] P. Seshu, "Finite Element Analysis", 2012 10th Printing.